

1. Tamer was simplifying the expression and made an error. His steps are shown. Find and fix the mistake.

$$\left(\frac{-2xy \cdot 3x^2}{5x^4y}\right)^2$$

Step 1: $\left(\frac{-6x^3y}{5x^4y}\right)^2$

Step 2: $\frac{-6x^6y^2}{5x^8y^2}$

Step 3: $\frac{-6}{5x^2}$

2. Write a note to Tamer explaining his error.

For questions 3-5, find an equivalent expression using the laws of exponents.

3. $\frac{-6a^7b^9}{(-2ab^3)^5}$

4. $\frac{(9x^8y) \cdot (-2x^2y^5)}{3x^5y^{10}}$

5. $\left(\frac{-3mn^3}{4m^4 \cdot 2m^2n^5}\right)^3$

6. Prove $2^{3x+5} = 32(8)^x$ by using the laws of exponents.

7. Prove $27^{6a-1} = \frac{(3^{6a})^3}{9 \cdot 3}$ by using the laws of exponents.

8. Write an equivalent expression for $(-5.2)^{2x^3} \cdot (-5.2)^{x^2} \cdot (-5.2)^{-5}$.

9. Use the law of exponents to write an equivalent expression for 4^{3-2c} .

10. Use the law of exponents to write an equivalent expression for 3.5^{3x+7} .